
UNIT 9 INFLATION AND MONETARY POLICY

Structure

- 9.0 Objectives
- 9.1 Introduction
- 9.2 Money
 - 9.2.1 What is Money?
 - 9.2.2 How Do We Measure Money?
- 9.3 Inflation
 - 9.3.1 What is Inflation?
 - 9.3.2 Inflation Measurement in India
 - 9.3.3 Costs of Inflation
- 9.4 Money and Prices
 - 9.4.1 Theoretical Perspectives on Money and Prices
 - 9.4.2 Causes of Inflation
- 9.5 Monetary Policy in India
 - 9.5.1 Objectives of Monetary Policy in India
 - 9.5.2 Instruments of Monetary Policy in India
 - 9.5.3 Monetary Policy Transmission
 - 9.5.4 Evolution of Monetary Policy in India
 - 9.5.5 Inflation Targeting Framework
- 9.6 Let Us Sum Up
- 9.7 Term-end Exercises
- 9.8 Key Words
- 9.9 References
- 9.10 Answers or Hints to Check Your Progress Exercises

9.0 OBJECTIVES

The purpose of this unit is to develop among learners an understanding of the framework and operating procedure of the monetary policy in India. After going through this unit, you will be able to:

- explain what money is and what inflation means and how money affects inflation;
- discuss what the instruments of monetary policy are;
- appreciate the evolution of monetary policy in India; and
- state the Inflation Targeting Framework in India.

9.1 INTRODUCTION

In this unit, we would be discussing inflation and monetary policy in India. Inflation is an economic issue that is a common topic of discussion for the public

as well as policymakers and researchers alike. It is an economic phenomenon that has a realisable impact on the daily lives of every citizen. You must have often heard your grandparents narrating stories about how, during their childhood, they were able to buy things for very small amounts of money compared to the present. They might remark that money has lost its value. Inflation, simply put, refers to the rate of increase in the prices of commodities during a given period. With a positive level of inflation, you are able to buy lesser amounts of goods and services using the same amount of money. Therefore, before we discuss inflation, it is important to understand what money is and how it affects the prices in the economy. We also discuss the monetary policy framework in India and see how it has evolved over the years, in sync with the requirements of the evolving economy.

9.2 MONEY

9.2.1 What is Money?

Money is something that we encounter in our daily lives. Despite this, it may be difficult for someone to define what money is or how it functions. Money is defined as anything that is generally acceptable as a medium of payment. Essentially, money serves three purposes:

- **It acts as a store of value:** Money can be a means to save a person's endowments, including labour, and used later for transactions. That is, income or wealth of individuals can be kept in the form of money and used for transactions in the future. It allows purchasing power to be taken forward from present to future. Thus, money acts as a store of value.
- **It acts as a unit of account:** Money acts as a common denominator on which all prices are measured. The value of goods and services exchanged in the economy is measured in terms of money.
- **It acts as a medium of exchange:** Since money has the above two features, it is generally acceptable as a means of payment for exchange of goods and services. Without money, any exchange in the economy would require a double coincidence of wants. That is, a transaction would take place only if both parties in the transaction had something that the other needed. With a generally acceptable means of payment, the need for double coincidence of wants is obviated by separating the transaction into two parts. First, the two parties engage in a transaction where one party gives money in exchange for the good or service he or she needs. In the second part of the transaction, the recipient of money in the first transaction uses the money to buy the goods or services that he or she requires.

Historically, many items have been used as money. Items with intrinsic value such as gold and silver were a form of money used for transactions in some societies. This form of money is called commodity money. Different societies have used everything from salt and grain to stones and precious metals as money at different points of time in history. Even in modern societies, commodities are known to have been used as a generally accepted means of payment. An interesting and often quoted example of commodity money is the use of cigarettes for transactions in Nazi prisoner of war camps during the second World War. The prevalent form of modern money is fiat money. Fiat money has no intrinsic value.

Its value is derived from a government order or a fiat. The coins and currency notes that we use in our daily lives are examples of fiat money.

9.2.2 How Do We Measure Money?

Any measure of the quantity of money should include the quantity of assets used in the economy for transactions. Currency notes and coins are obvious candidates for money. In addition to these, chequeable deposits may also be considered part of the money supply. In this manner, you may think of many other assets that could be considered for inclusion in the measure for money supply. It must therefore be clear to you that while defining money is a theoretical or conceptual matter, measuring money supply is an empirical question. RBI has the following measures of money supply:

Reserve Money (M0) = Currency in circulation + Bankers' deposits with the RBI
+ 'Other' deposits with the RBI

Narrow Money (M1) = Currency with the public + Demand deposits with the banking system + 'Other' deposits with the RBI

M2 = M1 + Savings deposits of post office savings banks

Broad Money (M3) = M1 + Time deposits with the banking system

M4 = M3 + All deposits with post office savings banks (excluding National Savings Certificates)

'Other' deposits with the RBI include deposits from foreign central banks, multilateral institutions, financial institutions and sundry deposits net of IMF Account No.1.

9.3 INFLATION

9.3.1 What is Inflation?

As you know, the price of a commodity refers to the amount of money you give in exchange for buying a unit of that commodity. The general price level refers to a measure of the overall price level in an economy. Inflation is the rate of increase in the general price level over a period of time. However, it is not straightforward to arrive at such a general price level, as in reality, such a measure does not exist. What we can do, at best, is to construct a measure that could be considered to represent this hypothetical value. To arrive at a single measure for the overall prices in the economy, we often look at a weighted average price of a basket of goods and services consumed by a representative consumer in the economy. This is achieved through the construction of a price index, which is a weighted average of price relatives of goods and services in a representative consumption basket. The changes in the price index over a given period reflect the changes in the prices of the goods and services in the consumption basket and therefore their growth rates are used as a measure of inflation.

9.3.2 Inflation Measurement in India

In India, two major types of price indices are used for calculation of inflation. These are the Consumer Price Index (CPI) and the Wholesale Price Index (WPI). As the name suggests, CPI reflects the average retail price level as faced by the consumers. There are four measures of CPI in India. CPI (Rural, Urban and

Combined), CPI (Industrial Workers), CPI (Rural Labourers) and CPI (Agricultural Labourers). Table 9.1 gives more details about the price indices used in India. Out of these, the most important one is CPI-Combined, being used for monetary policy purposes. CPI, separately for rural and urban areas, have been estimated for a longer duration in India. The base year for CPI (Industrial Workers) was recently updated to 2016 from the earlier base of 2001. On the other hand, WPI reflects the price level for the first point of wholesale or bulk transactions. In India, WPI is taken as a proxy for producers prices (i.e. prices faced by producers), which unlike other countries are not estimated. The current WPI series has a base year of 2011-12. It is also calculated based on Laspeyres price index formula.

Table 9.1: Different Price Indices in India

Name of the Index	Base Year	Organisation Compiling the Index
CPI (Rural, Urban and Combined)	2012	National Statistical Office
CPI (Industrial Workers)	2016	Labour Bureau
CPI (Agricultural Labourers)	1986-87	Labour Bureau
CPI (Rural Labourers)	1986-87	Labour Bureau
WPI	2011-12	Office of Economic Adviser, Department for Promotion of Industry and Internal Trade

9.3.3 Costs of Inflation

Inflation is generally thought of as imposing hardships on the citizens. High inflation is thought to make goods and services unaffordable. However, if incomes rise at a rate that is at least as much as the rate of inflation, the goods and services would continue to remain affordable. However, there may be other, not so obvious, economic costs of inflation. One cost associated with high inflation is one that arises from the fact that people may choose to hold lower cash with themselves and keep a larger amount in the banks as deposits. This would mean frequent visits to the bank to withdraw money for their daily transactions. The inconvenience of frequently visiting the bank is dubbed as shoe-leather cost of inflation. With input prices frequently changing during a high inflation scenario, firms have to change the prices of their products frequently. The cost associated with frequently changing the posted prices is called the menu cost of inflation. As such, if frequent price changes are not synchronised, it may lead to the introduction of relative price distortions leading to inefficient allocation of resources in the economy. High inflation also introduces uncertainty into economic decisions. Firms and households may find it difficult to differentiate between relative and general price changes. Further, high inflation may lead to undesirable redistribution of income in the economy. Low-income households are usually dependent on wage payments and cash holdings for their day-to-day transactions. During a case of high inflation, cash holdings lose their real values quickly. If wages are not indexed to inflation, as may be the case in the informal sector, it may take a while before wages respond to increased prices. On the other hand, richer households which hold wealth in non-cash assets would not be affected as much. This means that high levels of inflation may widen the gap between the rich and the poor in the economy.

9.4 MONEY AND PRICES

9.4.1 Theoretical Perspectives on Money and Prices

Depending on the school of thought followed, the theory of how money supply affects prices and other economic variables varies. In the classical perspective, the price level was determined by the quantity of money in the economy. This view, called the *Quantity Theory of Money*, which was later revived by the Monetarist school, has been a dominant explanation for the relationship between money and prices. We can express this in the form of the *equation of exchange*:

$$M \times V = P \times Y$$

Where, M is the quantity of money, V is the income velocity of money – the number of times a rupee is used in transactions, that is the number of times on an average a rupee changes hand, P is the price level and Y is the output produced in the economy. Output being supply determined in the Classical view and the velocity of money determined by payment technology and institutional factors, being unchanged in the short-run, this identity says that an increase in the quantity of money would lead to a proportionate increase in the prices. Thus, with these assumptions, the equation of exchange translates into the quantity theory of money, explaining that the price level is determined by the quantity of money.

The equation of exchange can be represented in terms of growth rates by taking logs on both sides and differentiating with respect to time. We get,

$$\frac{1}{M} \frac{dM}{dt} + \frac{1}{V} \frac{dV}{dt} = \frac{1}{P} \frac{dP}{dt} + \frac{1}{Y} \frac{dY}{dt}$$

Which is nothing but,

Growth rate of money supply + Growth rate of velocity of money =
Inflation + Growth rate of output

Rearranging, we get,

Inflation = Growth rate of money supply + Growth rate of velocity of money –
Growth rate of output

With a stable velocity of money, growth in it is zero. Output is dependent on the factors of production. It is then straightforward that in the quantity theory of money, the inflation level is under the ultimate control of the monetary authority through the control on the quantity of money. The monetary authority can vary the inflation rate under this framework by varying the growth rate of money.

In the Keynesian perspective, money affects income via interest rates. An increase in money supply lowers the interest rate and stimulates aggregate demand and income. In an economy with excess capacity, higher aggregate demand should only lead to use of the earlier unused capacity in the economy and therefore should not lead to inflation.

In the late 1950s-60s, the Phillips Curve relationship came into the limelight, which provided a link between inflation and unemployment. According to this relationship, in the short-run there is a tradeoff between inflation and unemployment: an economy could achieve a lower unemployment rate by

stimulating demand in the economy at least in the short-run if it was willing to tolerate a higher level of inflation. In the long-run, however, this relationship is expected to breakdown as workers would want their wages adjusted to the higher rates of inflation and therefore, as a consequence, unemployment again rising.

9.4.2 Causes of Inflation

As we have seen above, the quantity of money is an important determinant of the level of inflation in the economy. The monetarist, Milton Friedman stated, “Inflation is always and everywhere a monetary phenomenon.” There is considerable agreement that, in the long-run money supply growth can be seen as a major determinant of inflation. In the short-run, however, fluctuations in the demand and supply conditions can have a major impact on the movement in the level of prices. Higher government spending, for example, can raise the level of aggregate demand in the economy and therefore put upward pressure on prices. This is an example of ‘demand-pull’ inflation. Another cause of inflation is increase in the costs of production due to increases in the costs of raw materials and other inputs. This is called ‘cost-push’ inflation. Sudden changes in supply conditions can alter the level of prices of the commodities concerned. You might have observed that years with deficient rainfall, prices of agricultural commodities witness sharp spikes. Insufficient rainfall affects production of agricultural commodities and therefore may lead to shortfall in the supply of these commodities, thus affecting the prices. These supply side factors are not generally expected to be affected by monetary policy.

For economic analysis, overall inflation (also called headline inflation) is usually divided into a “Core” part and a non-core part. Core inflation refers to that component of inflation that is long-term in nature. That is, Core inflation is calculated by separating out the volatile components of inflation that cause fluctuations in the short-term. These, in the context of a developing economy like ours, are usually components like food items that depend on seasonal and weather factors to determine their supply or fuel items that may be influenced by supply side shocks in the international markets. One would expect Core inflation to be more responsive to monetary policy than non-core inflation.

Check Your Progress 1

- 1) Consider an economy with a constant velocity of money. Money supply grows by 10 per cent per year and real income grows by 5 per cent per year. As per the Quantity Theory of Money that you have learnt above, what is the average rate of inflation?

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- 2) Download the yearly data on M3 (from average monetary aggregates) and nominal GDP of India (GDP at current prices) for years 2004-05 onwards from RBI’s Database on Indian Economy (<https://dbie.rbi.org.in/>). Now for each year, calculate the velocity of money by dividing the nominal GDP

for each year by the corresponding M3 number. Plot this as a line chart for these years. What trend do you observe in the velocity of money? What do you think are the reasons for the observed trend? What do you think would be the effect of greater number of digital transactions on the velocity of money?

9.5 MONETARY POLICY IN INDIA

9.5.1 Objectives of Monetary Policy in India

As the monetary authority of the country, the Reserve Bank of India formulates, implements, and monitors the monetary policy of the country. As per the Preamble of the Reserve Bank of India Act 1934, the primary objective of monetary policy is to maintain price stability while keeping in mind the objective of growth. This is a direct consequence of its mandated function to “regulate the issue of Bank notes and keeping of reserves with a view to securing monetary stability in India and generally to operate the currency and credit system of the country to its advantage”. As we have seen, maintaining price stability is crucially dependent on the money supply in the economy. A high level of inflation may have an adverse effect on growth. Therefore, the objective is to maintain a low and stable level of inflation in the economy. Further, achieving high growth in the economy requires that credit flows to the most productive sectors of the economy.

9.5.2 Instruments of Monetary Policy in India

As the monetary policy framework has evolved over the years, the operating procedure of RBI's monetary policy has also evolved. RBI relies on a set of direct as well as indirect policy instruments to conduct its monetary policy. During the initial years, the focus of RBI was to regulate the demand and supply of credit through bank rate, reserve requirements and Open Market Operations.

As we have seen earlier, the need to finance government deficit dominated the monetary policy actions during the period between the 1950s and 1970s. Selective credit control mechanisms were used to channel credit to priority sectors. Between 1970 and 1990, the Statutory Liquidity Ratio (SLR), the share of net demand and time liabilities (NDTL) that a bank is required to maintain in safe and liquid assets, such as, unencumbered government securities, cash and gold, was increased several times. Within the period of 1980's, with a view to regulating liquidity and to provide resources for vital public sector investment within the framework of national priorities without generating reserve money, it was decided to increase the SLR. During the 1980s, a number of money market instruments such as inter-bank participation certificates (IBPCs), certificates of deposit (CDs) and Commercial Paper (CP) were also introduced.

Another important monetary policy instrument during the period was the Cash Reserve Ratio (CRR). The Cash Reserve Ratio (CRR) refers to the average daily balance that a bank is required to maintain with the RBI as a percentage of its Net demand and time liabilities (NDTL) that the RBI may notify from time to time in the Gazette of India. If the average daily balance held at RBI by a scheduled bank during any fortnight is below the minimum prescribed, the bank shall be liable to pay a penal interest on the amount of shortfall. CRR as a monetary policy instrument can be used to influence the money supply by the monetary

authority. An increase in CRR would lead the banks to have lesser amounts of money with them for lending with the same amount of deposits. This would reduce the amount of credit created and therefore reduce the money supply through the money multiplier process. In the 1980's the CRR was changed in different time periods as one of the important policy measures.

With liberalisation in the Indian economy and the financial sector reforms, there was a shift from direct to indirect, market based policy instruments. In the period 1990-91, the Narasimham committee had recommended that the CRR should continue to be used as an instrument of monetary control. But it was recognised that the existing levels were high and needed to be brought down. The year 1996-97 was significant in many ways in relation to the conduct of monetary policy in India. It was the first time the CRR was reduced sharply by 4 percentage points within a financial year in the five phases. The CRR was brought down to 9.5 per cent and SLR to 25 per cent by 1997. The process of deregulation of interest rates was also initiated in the early 1990s simplifying the complex structure of deposit and lending interest rates.

Following the Narasimham Committee-II Report (1998), the Interim Liquidity Adjustment Facility (ILAF) was introduced in April 1999, to determine the corridor for money-market rates. With subsequent revisions, the revised Liquidity Adjustment Facility (LAF) was announced in October 2004 and has now emerged as the principal operating instrument of monetary policy. The LAF consists of overnight as well as term repo auctions. Repurchase agreements, called repo (repurchase options) or reverse repo are agreements in which two parties agree to sell and repurchase the same security. They are usually used for overnight borrowing. Under a repurchase agreement the seller sells specified securities with an agreement to repurchase the same at a mutually decided future date and price. Similarly, the buyer purchases the securities with an agreement to resell the same to the seller on an agreed date at a predetermined price. Such a transaction is called a repo when viewed from the perspective of the seller of the securities and reverse repo when viewed from the perspective of the buyer of the securities. Under the LAF, the repo rate is the rate at which the Reserve Bank provides overnight liquidity to banks against the collateral of government and other approved securities under the liquidity adjustment facility (LAF). On the other hand, the rate at which the Reserve Bank absorbs liquidity, on an overnight basis, from banks against the collateral of eligible government securities under the LAF is called the reverse repo rate.

With the LAF, repo rate has become the single independent policy rate. The reverse repo rate is placed below the repo rate. Under the LAF, a Marginal Standing Facility (MSF) was also instituted. MSF is a facility under which scheduled commercial banks can borrow additional amounts of overnight money from the RBI at a penal rate of interest. The MSF rate and reverse repo rate determine the corridor for the daily movement in the weighted average call money rate with the repo rate in the middle of the corridor (Figure 9.1). The LAF operates through daily repo and reverse repo auctions thereby setting a corridor for the short-term interest rate consistent with the policy objectives.

In the current monetary operating framework, reliance on direct instruments of monetary policy has been reduced and liquidity management in the system is carried out through open market operations (OMO) in the form of outright purchases/sales of government securities and repo and reverse repo operations

under Liquidity Adjustment Facility (LAF). The LAF enables RBI to modulate short-term liquidity under varied financial market conditions in order to ensure stable conditions in the overnight (call) money market.

Bank rate is the rate at which RBI is ready to buy or rediscount bills of exchange or other commercial papers. The Bank Rate is published under Section 49 of the Reserve Bank of India Act, 1934. This rate has been aligned to the MSF rate and, therefore, changes automatically as and when the MSF rate changes alongside policy repo rate changes.

Market Stabilisation Scheme (MSS) is an instrument for monetary management was introduced in 2004. Surplus liquidity of a more enduring nature arising from large capital inflows is absorbed through the sale of short-dated government securities and treasury bills. The cash so mobilised is held in a separate government account with the Reserve Bank.

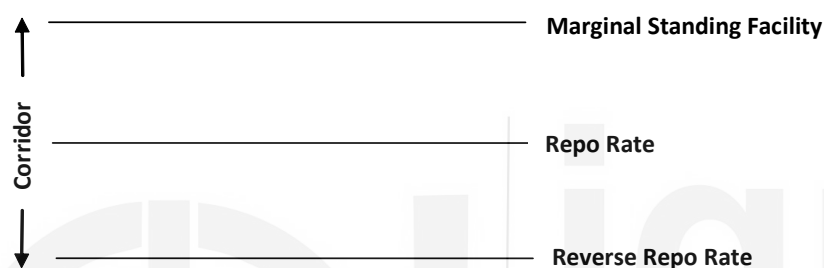


Fig. 9.1: The Liquidity Adjustment Facility Framework

9.5.3 Monetary Policy Transmission

Monetary policy transmission is the process through which monetary policy affects the ultimate policy target. That is, it is the process through which a change in the policy rates by RBI, through changes in lending rates of banks, deposit rates, asset prices and economic activities, ultimately affects the inflation rate. As such, it is a complex process involving uncertainty in the channel, quantum, and timing of impact.

Monetary policy acts through influencing the cost and availability of credit. A change in RBI's policy rate, through its impact on the short-term interest rates and further on rates of longer-term instruments, affects the cost of funds for banks. This would affect the rate at which banks would be willing to lend as well as the rate that they would be willing to offer their depositors. For example, a reduction in the policy rate leads to a lower cost of funds for the banks and therefore lower lending rates. This would induce households and businesses to avail greater credit from the banks and thus boost the aggregate demand in the economy. The higher demand in the economy would put upward pressure on wages and prices. Lower interest rates also boost the demand for assets and thus increasing their prices. Higher asset prices imply greater wealth for the asset owners and therefore may lead to greater spending in the economy, thus again boosting the demand and pushing overall prices up. Lower interest rates could also impact inflation through the exchange rate channel. Lower interest rates imply lower returns for foreign investors and therefore may lead to capital flight from the country. This would cause the currency to depreciate. This means that imports are much more costlier now compared to earlier, causing prices to go up.

The effectiveness of monetary policy essentially depends on the institutional framework available for transmitting impulses released by the central bank. It

has been observed that monetary policy transmission in India has been slow and generally incomplete. India has a financial system dominated by commercial banks. A major part of the total liabilities is in the form of deposits. A large number of deposits have maturity over one year meaning that the changes in policy rates would only impact new deposits. High NPAs have also impacted the profitability and liquidity of the banks affecting the transmission process.

9.5.4 Evolution of Monetary Policy in India

The approach to monetary policy has been evolving over the years as per the requirements of a dynamic economy since the inception of RBI as the country's central bank in the year 1935. Overall price stability and growth have been the broad guiding principles of monetary policy. RBI was nationalised in 1949 and Banking Regulation Act was enacted the same year. The latter required the banks to maintain a Statutory Liquidity Ratio (SLR) as a percentage of their net time and demand liabilities in the form of cash, gold or approved securities (Mohanty, 2017). During the initial decades after independence of the country, the role of RBI was to support the developmental efforts of the government by ensuring adequate flow of credit to the priority sectors of the economy. Owing to the primacy of public investment in the development policy during this period, there was persistent deficit financing. Issuance of ad-hoc treasury bills by the government leading to automatic monetisation of deficits, as well as ensuring flow of funds from banks through higher SLR were the main modes of financing Government deficits. The monetisation of deficits, war with Pakistan in 1971, oil price shocks during 1973 and 1979, as well as droughts in 1973 and 1979 led to high inflation during the 1970s (Mohanty, 2017). Inflation increased from 10 per cent in 1972-73 to 25 per cent in 1974-75.

The Sukhamoy Chakravarty Committee formed in 1982 to assess the working of the monetary system, recommended primacy of price stability in monetary policy. Chakravarty Committee recommended a monetary targeting framework. Between the 1980s and late 1990s, a monetary targeting framework was followed by the RBI for monetary policy. Development and price stability continued to be the primary objectives of monetary policy. Under this framework, broad money (M3) was the intermediate target of monetary policy.

As you would know, the 1990s was a decade of many broad-based and transformative reforms in the Indian economy. Due to the developments and reforms in the financial sector, indirect instruments such as interest rates had started gaining effectiveness as transmitters of policy signals. Liberalisation in the financial markets made money demand unstable. Further, due to the greater integration of the Indian economy with the global economy and the ensuing inflow of foreign capital made it difficult to completely control the money supply. Due to these reasons, RBI shifted to a "Multiple Indicators" approach in 1998-99. Under this approach, Monetary policy was formulated based on a wide set of variables and more emphasis was put on the rates channel for monetary policy (Mohanty, 2011).

In September 2013, RBI set up an expert committee headed by its then Deputy Governor, Dr. Urjit Patel, with a view to strengthen the monetary policy framework while among other things, making it transparent and predictable. It recommended that inflation should be set as the nominal anchor for monetary policy framework and that it should be set as the predominant objective of